Probabilistic Reasoning

Uncertainty

It refers to situations where there is a lack of complete information or knowledge about a particular aspect, leading to ambiguity and unpredictability.

Why do we need reasoning under uncertainty?

- There are many situations where uncertainty arises:
  - ✓ When you travel you reason about the possibility of delays
  - ✓ When an insurance company offers a policy it has calculated the risk that you will claim
  - ✓ When your brain estimates what an object is it filters random noise and fills in missing details
  - ✓ When you play a game you cannot be certain what the other player will do
  - ✓ A medical expert system that diagnoses disease has to deal with the results of tests that are sometimes incorrect
- Systems which can reason about the effects of uncertainty should do better than those that don't.
- But how should uncertainty be represented?

Example

<u>Example:</u> Drawing Cards from a Deck

✓ Calculate the probability of drawing a red card (heart or diamond) from the deck? → 26/52 = ½ = 0.5

- Now, let's introduce a <u>condition</u>: you want to calculate the probability of drawing a diamond given that you've already drawn a red card.

<u>**Let's represent the events:**</u>

A: Drawing a diamond (event of interest).

B: Drawing a red card (condition).

Calculate **P(A | B)**, which means "**the probability of drawing a diamond given that we've drawn a red card**."

$P(A \cap B) = 13/52 = .25$

$P(A \mid B) = P(A \cap B) / P(B) = = 0.25 / 0.5$

Reasoning under uncertainty

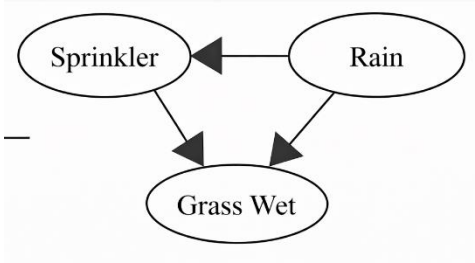
- How to build network models to reason under uncertainty according to the laws of probability theory?

- Various types of reasoning strategies used to handle uncertainty
  1. Probabilistic Models
  2. Bayesian Networks
  3. Monte Carlo Methods
  4. Decision Theory
  5. Fuzzy Logic
  6. Qualitative Reasoning

Bayesian Networks

- Belief network/probabilistic network/causal network /knowledge map.
- Bayesian Nets (BN) are a network-based framework for representing and analyzing models involving uncertainty.
- It is probabilistic graphical model that represents a set of variables and their conditional dependencies via a directed acyclic graph (DAG).
- Show how they can be used to capture uncertain knowledge in a natural & efficient way.

Definition of a Bayesian Network



Knowledge structure:

- variables are nodes
- arcs represent probabilistic dependence between variables
- conditional probabilities encode the strength of the dependencies

Computational architecture:

- computes posterior probabilities given evidence about some nodes

Bayesian networks

- A simple, graphical notation for conditional independence assertions & hence for compact specification of full joint distributions
- A Bayesian network is a directed graph in which each node is annotated with quantitative probability information.

Syntax:

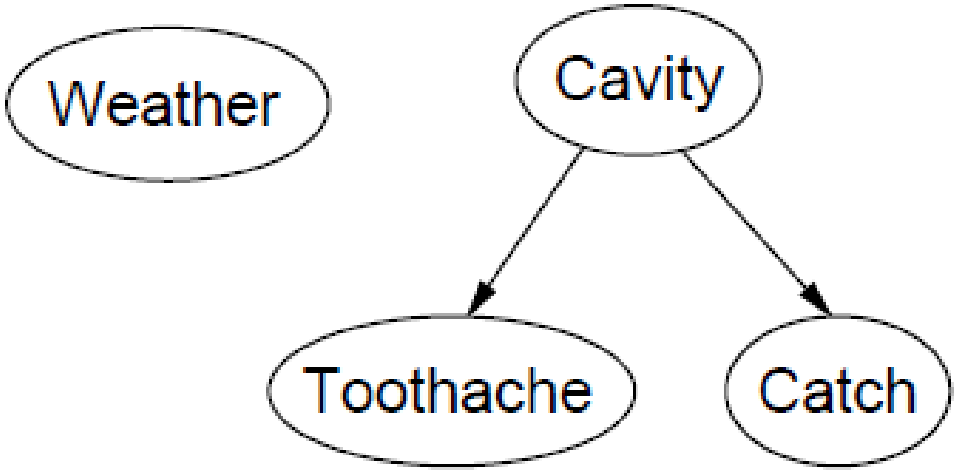
- A set of nodes, one per variable
- A set of directed links/arrows connects pairs of nodes. If there is an arrow from node X to node Y, X is said to be a *parent* of Y
- A directed, acyclic graph (link ≈ "directly influences")
- A conditional distribution for each node given its parents:

$$P(X_i|Parents(X_i))$$

- In the simplest case, conditional distribution represented as a conditional probability table (CPT) giving the distribution over X_i for each combination of parent values

Example

Topology of network encodes conditional independence assertions:



$Weather$ is independent of the other variables

$Toothache$ and $Catch$ are conditionally independent given $Cavity$

Cavity is a direct cause of *Toothache* & *Catch*, whereas no direct causal relationship exists between *Toothache & Catch*

Another Example

- You have a new burglar alarm installed at home. It is fairly reliable at detecting a burglary, but also responds on occasion to minor earthquakes.
- You also have 02 neighbors, John & Mary, who have promised to call you at work when they hear the alarm.
- John always call hears the alarm, but sometimes confuses the telephone ringing with the alarm & calls then, too.
- Mary, on the other hand, likes rather loud music & sometimes misses the alarm altogether.
- Given the evidence who has or has not called,

-estimate the probability of a burglary

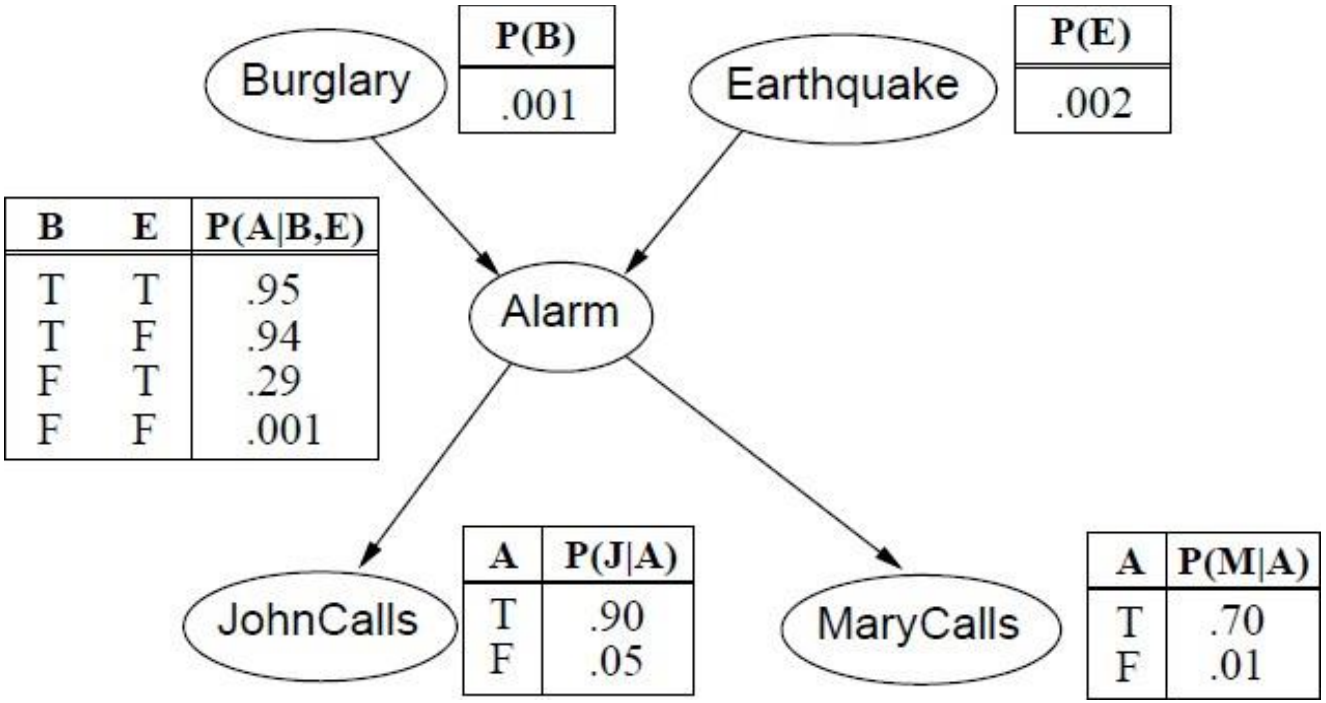
Example

I'm at work, neighbor John calls to say my alarm is ringing, but neighbor Mary doesn't call. Sometimes it's set off by minor earthquakes. Is there a burglar?

Variables: $Burglar$, $Earthquake$, $Alarm$, $JohnCalls$, $MaryCalls$
Network topology reflects "causal" knowledge:

- A burglar can set the alarm off
- An earthquake can set the alarm off
- The alarm can cause Mary to call
- The alarm can cause John to call

Example Question



QS - John calls (J) and Mary calls (M). What is the probability that neither burglary nor earthquake occurred if the alarm rang?

Solution

- The example given calculates the probability of a specific scenario: **John calls (J), Mary calls (M), Alarm rings (A)**, and **neither Burglary (B) nor Earthquake (E) occurs.**
- We're interested in calculating:

$$P(j, m, a, \neg b, \neg e)$$

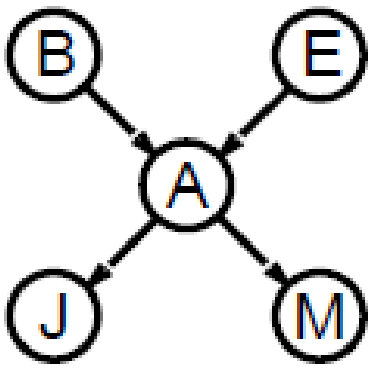
- This expression expands as follows, using conditional probabilities for each variable given its parents:

$$P(j, m, a, \neg b, \neg e) = P(j|a) \cdot P(m|a) \cdot P(a|\neg b, \neg e) \cdot P(\neg b) \cdot P(\neg e)$$

Global semantics

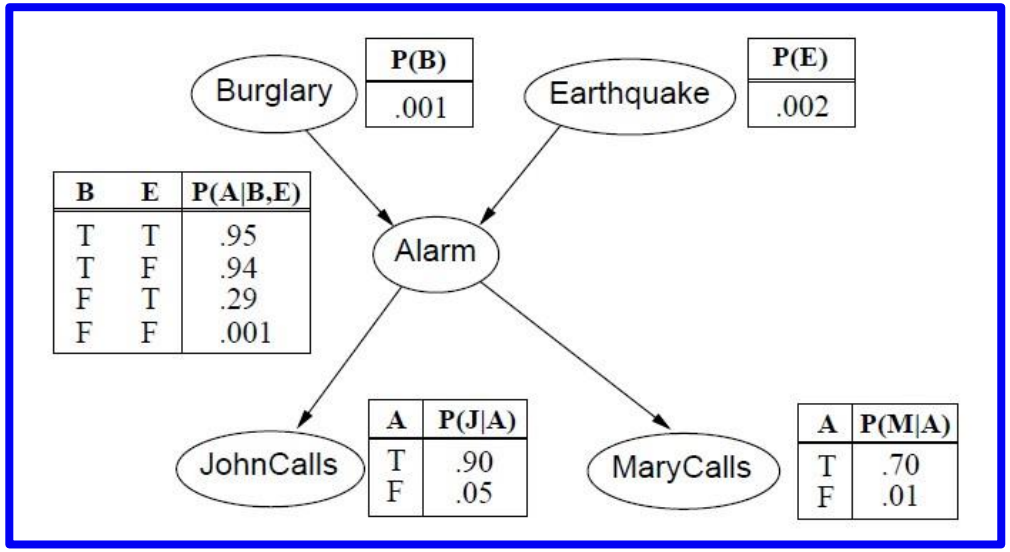
"Global" semantics defines the full joint distribution as the product of the local conditional distributions:

$$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P(x_i | parents(X_i))$$



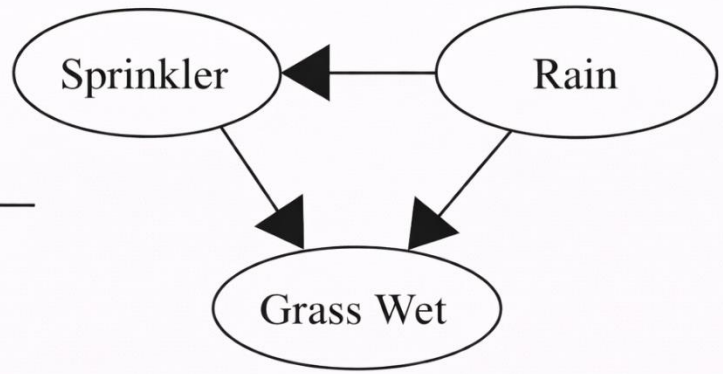
e.g., $P(j \wedge m \wedge a \wedge \neg b \wedge \neg e)$

$$= P(j|a)P(m|a)P(a|\neg b, \neg e)P(\neg b)P(\neg e)$$
$$= 0.9 \times 0.7 \times 0.001 \times 0.999 \times 0.998$$
$$\approx 0.00063$$



John calls (J) and Mary calls (M). What is the probability that neither burglary nor earthquake occurred if the alarm rang?

Example – "Rain – Sprinkler – Grass Wet"



Sprinkler

Rain	S
F	0.4
T	0.01

Rain

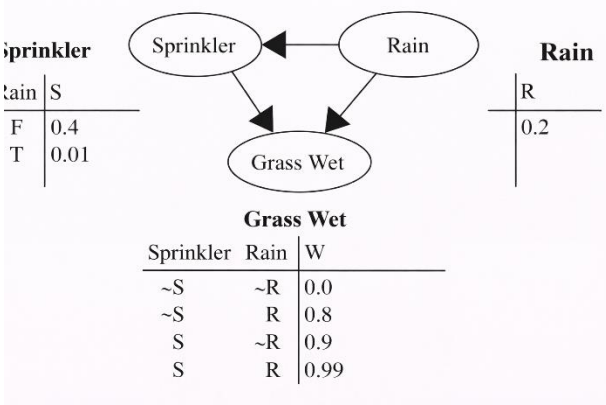
R
0.2

Grass Wet

Sprinkler	Rain	W
~S	~R	0.0
~S	R	0.8
S	~R	0.9
S	R	0.99

Given that the grass is wet (W), what is the probability that it is due to both rain (R) and sprinkler (S)?

Solution



<u>1) Network structure</u>

From the image:

- ❑ **Rain (R)** affects **Sprinkler (S)**
- ❑ **Rain (R)** and **Sprinkler (S)** both affect **Grass Wet (W)**

So the Bayesian network is:

$$R \rightarrow S, R \rightarrow W, S \rightarrow W$$

<u>2) Given probabilities from the image</u>

A. Prior probability of Rain

$$P(R) = 0.2, \qquad P(\neg R) = 0.8$$

B. Conditional probability of Sprinkler given Rain

From the table:

- ❑ When **Rain = False**:

$$P(S \mid \neg R) = 0.4, \qquad P(\neg S \mid \neg R) = 0.6$$

- ❑ When **Rain = True**:

$$P(S \mid R) = 0.01, \qquad P(\neg S \mid R) = 0.99$$

Solution

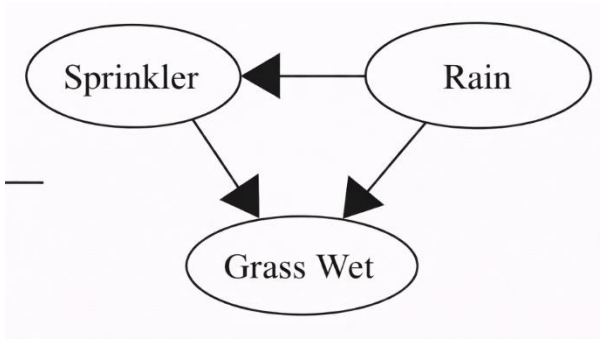
C. <u>Conditional probability of Grass Wet</u>

From the table:

$$P(W \mid \neg S, \neg R) = 0.0, \qquad P(\neg W \mid \neg S, \neg R) = 1.0$$
$$P(W \mid \neg S, R) = 0.8, \qquad P(\neg W \mid \neg S, R) = 0.2$$
$$P(W \mid S, \neg R) = 0.9, \qquad P(\neg W \mid S, \neg R) = 0.1$$
$$P(W \mid S, R) = 0.99, \qquad P(\neg W \mid S, R) = 0.01$$

Sprinkler

Rain	S	$\neg S$
F	0.4	0.6
T	0.01	0.99



Rain

	R	$\neg R$
	0.2	0.8

Grass Wet

Sprinkler	Rain	W	$\neg W$
$\neg S$	$\neg R$	0.0	1.0
$\neg S$	R	0.8	0.2
S	$\neg R$	0.9	0.1
S	R	0.99	0.01

Solution

3) What we have to find

We need to find:

$$P(R\ AND\ S \mid W)$$

or,

$$P(\ R, S \mid W\ )$$

That means:

Given that the **grass is wet**, what is the probability that it is due to **both rain and sprinkler**?

4) Formula used

<u>Using conditional probability:</u>

$$P(R, S \mid W) = \frac{P(W,R,S)}{P(W)} \quad \text{[From, } \boldsymbol{P(A|B) = \frac{P(A \wedge B)}{P(B)}}\text{]}$$

So,

$$P(R, S \mid W) = \frac{P(W \mid R, S) \cdot P(S \mid R) \cdot P(R)}{P(W)}$$

Solution

5) Calculate the numerator

From the tables:

$$P(W \mid R, S) = 0.99$$
$$P(S \mid R) = 0.01$$
$$P(R) = 0.2$$

$So,$

$$P(W, R, S) = 0.99 \times 0.01 \times 0.2 = 0.00198$$

6) Calculate the denominator P(W)P(W)P(W)

To find P(W), we consider all possible cases (4 cases):

$$P(W) = P(W \mid S, R)P(S \mid R)P(R)$$
$$+P(W \mid \neg S, R)P(\neg S \mid R)P(R)$$
$$+P(W \mid S, \neg R)P(S \mid \neg R)P(\neg R)$$
$$+P(W \mid \neg S, \neg R)P(\neg S \mid \neg R)P(\neg R)$$

Now substitute the values:

$$P(W)$$
$$= 0.99(0.01)(0.2) + 0.8(0.99)(0.2) + 0.9(0.4)(0.8)$$
$$+ 0.0(0.6)(0.8)$$

Solution

Now calculate each term:

Case 1: S,R

$$0.99 \times 0.01 \times 0.2 = 0.00198$$

Case 2: ¬S,R

$$0.8 \times 0.99 \times 0.2 = 0.1584$$

Case 3: S,¬R

$$0.9 \times 0.4 \times 0.8 = 0.288$$

Case 4: ¬S,¬R

$$0.0 \times 0.6 \times 0.8 = 0$$

Now add them:

$$P(W) = 0.00198 + 0.1584 + 0.288 + 0$$
$$= 0.44838$$

7) Final calculation

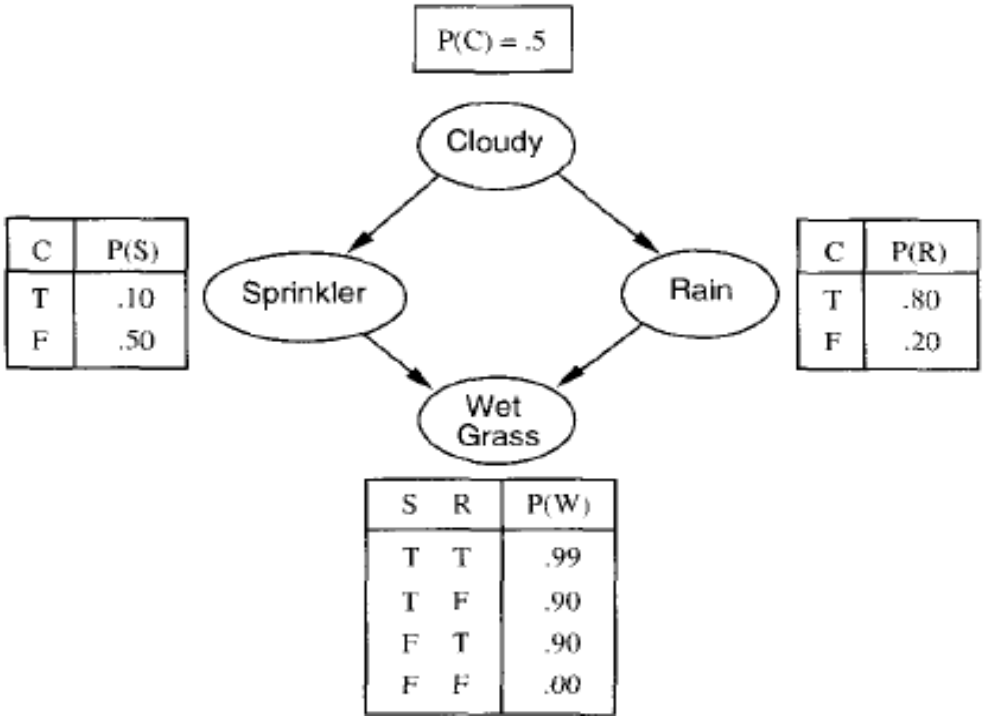
$$P(\,R, S \mid W\,) = \frac{0.00198}{0.44838} \approx 0.00442$$

$So,$

$P(R\ AND\ S \mid W) \approx 0.00442 \approx$ ***0.442%(Probability)***

Example

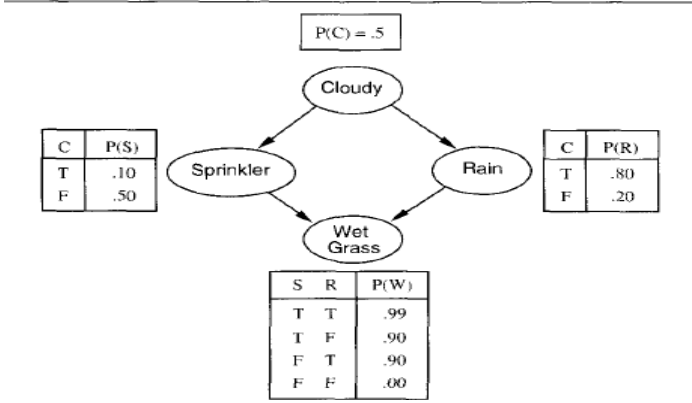
("Cloudy – Sprinkler – Rain – Wet Grass")



A multiply connected network with conditional probability tables.

Given that the grass is wet (W), what is the probability that it is due to rain (R) and not the sprinkler (S), assuming it is a cloudy day (C)?

Solution



1) Network structure

From the figure:

$$C \rightarrow S, C \rightarrow R, S \rightarrow W, R \rightarrow W$$

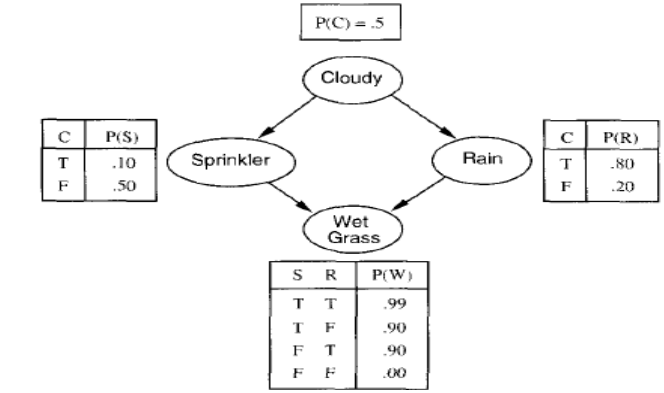
Where,

- C = Cloudy
- S = Sprinkler
- R = Rain
- W = Wet Grass

So:

- **Cloudy** affects both **Sprinkler** and **Rain**
- **Wet Grass** depends on both **Sprinkler** and **Rain**

Solution



2) Given probabilities

A. Prior probability of Cloudy

$$P(C) = 0.5,\ P(\neg C) = 0.5$$

Since the question says **assuming it is a cloudy day**, we use:

$$C = True$$

B. Conditional probability of Sprinkler given Cloudy

$$P(S \mid C) = 0.10, \qquad P(\neg S \mid C) = 0.90$$
$$P(S \mid \neg C) = 0.50, \qquad P(\neg S \mid \neg C) = 0.50$$

C. Conditional probability of Rain given Cloudy

$$P(R \mid C) = 0.80, \qquad P(\neg R \mid C) = 0.20$$
$$P(R \mid \neg C) = 0.20, \qquad P(\neg R \mid \neg C) = 0.80$$

D. Conditional probability of Wet Grass

$$P(W \mid S, R) = 0.99, \qquad P(\neg W \mid S, R) = 0.01$$
$$P(W \mid S, \neg R) = 0.90, \qquad P(\neg W \mid S, \neg R) = 0.10$$
$$P(W \mid \neg S, R) = 0.90, \qquad P(\neg W \mid \neg S, R) = 0.10$$
$$P(W \mid \neg S, \neg R) = 0.00, \qquad P(\neg W \mid \neg S, \neg R) = 1.00$$

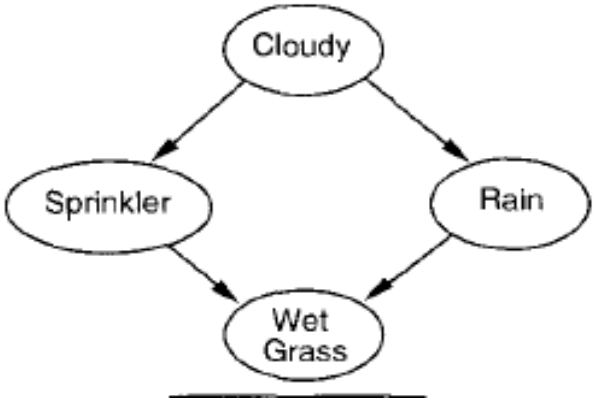
Solution

Cloudy

P(C)	$P(\neg C)$
0.2	0.8

Sprinkler

Cloudy	P(S)	$P(\neg S)$
T	0.10	0.90
F	0.50	0.50



Rain

Cloudy	P(R)	$P(\neg R)$
T	0.80	0.20
F	0.20	0.80

Grass Wet

Sprinkler	Rain	P(W)	$P(\neg W)$
T	T	0.99	0.01
T	F	0.90	0.10
F	T	0.90	0.10
F	F	0.00	1.0

Solution

We need to find:

$$P(R, \neg S \mid \overbrace{W, C}^{\text{Given That}}) = ?$$

That means:
Given that the **grass is wet** and **it is a cloudy day**, what is the probability that the wet grass is due to **rain and not the sprinkler**?

4) Formula used
Using conditional probability:

$$P(R, \neg S \mid W, C) = \frac{P(R \wedge \neg S \wedge W \wedge C)}{P(W, C)} \quad \textbf{[From, } \boldsymbol{P(A|B) = \frac{P(A \wedge B)}{P(B)}}\textbf{]}$$

$$= \frac{P(R, \neg S, W, C)}{P(W, C)}$$

Now, in this network:

- ✓ W depends only on S and R
- ✓ S and R are conditionally independent given C

So,

$$P(R, \neg S \mid W, C) = \frac{P(W \mid R, \neg S) \cdot P(R \mid C) \cdot P(\neg S \mid C) \cdot P(C)}{P(W \mid C) \cdot P(C)}$$

Solution

$$P(R, \neg S \mid W, C) = \frac{P(W \mid R, \neg S) \cdot P(R \mid C) \cdot P(\neg S \mid C)}{P(W \mid C)}$$

5) Calculate the numerator

$$P(W \mid R, \neg S) = 0.90$$
$$P(R \mid C) = 0.80$$
$$P(\neg S \mid C) = 0.90$$

$So,$

$$Numerator = 0.90 \times 0.80 \times 0.90 = 0.648$$

6) Calculate the denominator $\boldsymbol{P(W \mid C)}$

We must consider all possible ways the grass can be wet when C is true:

$$P(W \mid C) = \sum_{S,R} \mathrm{P(W \mid S, R) \cdot P(S \mid C) \cdot P(R \mid C)}$$

Solution

That is,

$$
\begin{aligned}
&P(W \mid C) \\
&= P(W \mid S, R)P(S \mid C)P(R \mid C) \\
&+ P(W \mid S, \neg R)P(S \mid C)P(\neg R \mid C) \\
&+ P(W \mid \neg S, R)P(\neg S \mid C)P(R \mid C) \\
&+ P(W \mid \neg S, \neg R)P(\neg S \mid C)P(\neg R \mid C)
\end{aligned}
$$

Now substitute the values:

$$
\begin{aligned}
&P(W \mid C) \\
&= 0.99(0.10)(0.80) + 0.90(0.10)(0.20) + 0.90(0.90)(0.80) \\
&+ 0.00(0.90)(0.20) \\
&= 0.0792 + 0.018 + 0.648 + 0 = 0.7452
\end{aligned}
$$

7) Final probability

Now,

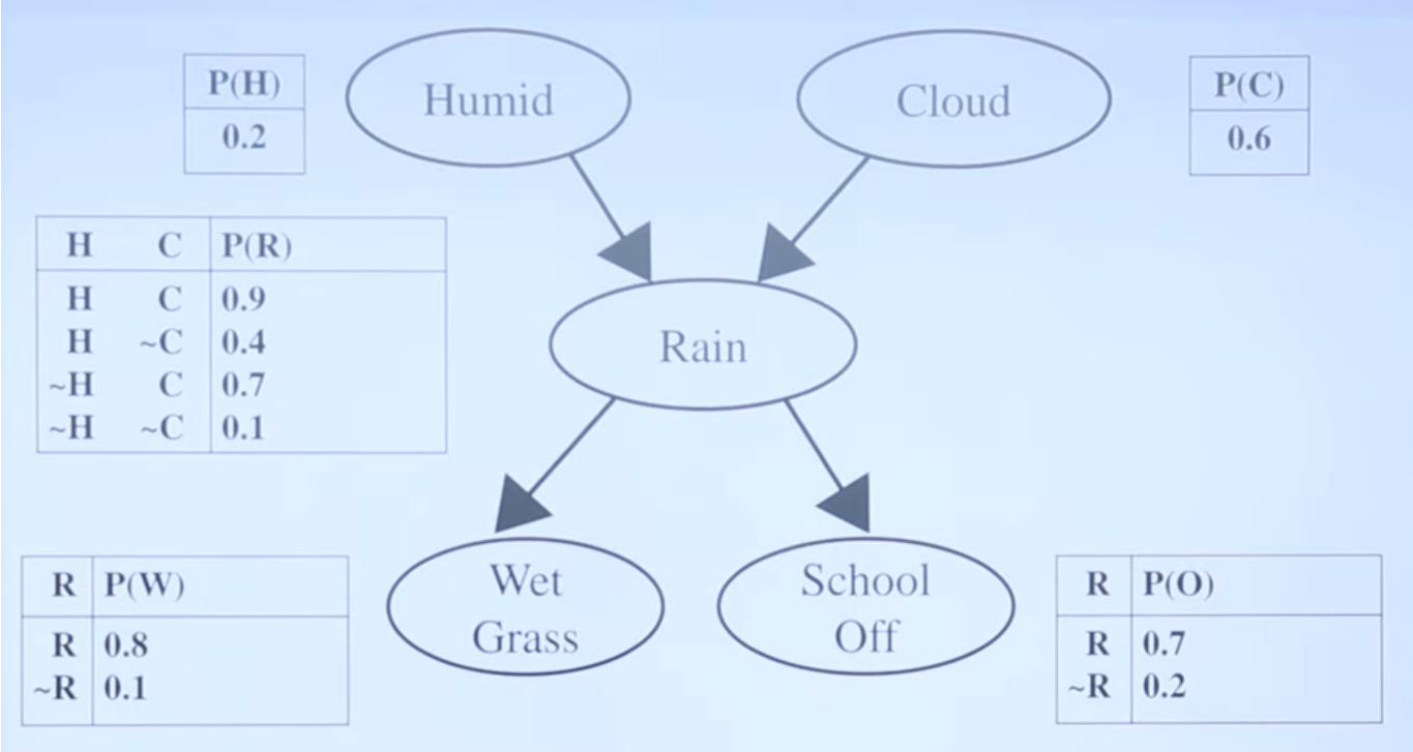
$$P(R, \neg S \mid W, C) = \frac{0.648}{0.7452}$$

$$P(R, \neg S \mid W, C) \approx 0.8696$$

So, the probability is -

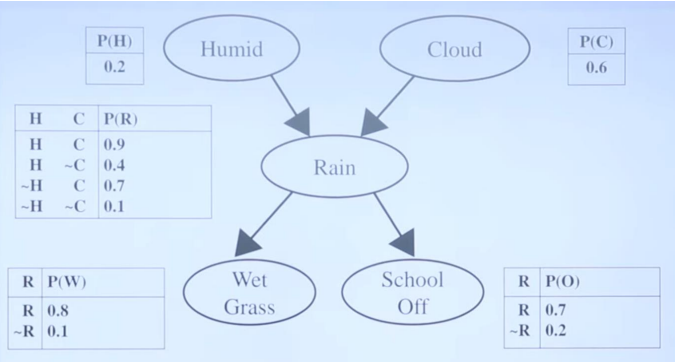
$$P(R, \neg S \mid W, C) \approx 86.96\%$$

Example



Using the Bayesian Network shown in the figure, calculate the joint probability that the school is off, it is raining, the weather is not humid, and it is cloudy.

Solution



1) Network structure

From the image, the Bayesian network is:

$$H \rightarrow R, C \rightarrow R, R \rightarrow W, R \rightarrow O$$

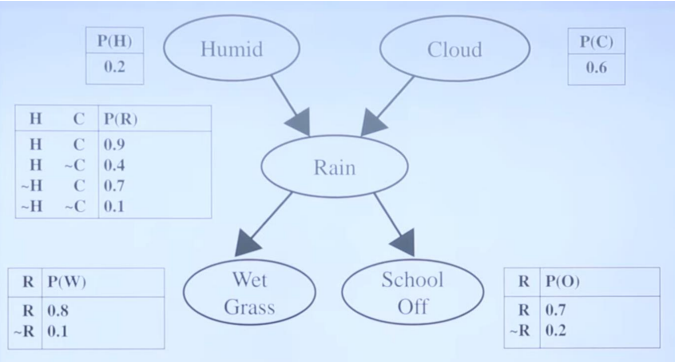
Where,

- $H = Humid$
- $C = Cloud$
- $R = Rain$
- $W = Wet\ Grass$
- $O = School\ Off$

So:

- ✓ **Humid** and **Cloud** influence **Rain**
- ✓ **Rain** influences **Wet Grass**
- ✓ **Rain** also influences **School Off**

Solution



2) Given probabilities

A. Prior probabilities

For **Humid**:

$$P(H) = 0.2, \qquad P(\neg H) = 0.8$$

For **Cloud**:

$$P(C) = 0.6, \qquad P(\neg C) = 0.4$$

B. Conditional probability of Rain

From the table:

$$P(R \mid H, C) = 0.9, \qquad P(\neg R \mid H, C) = 0.1$$
$$P(R \mid H, \neg C) = 0.4, \qquad P(\neg R \mid H, \neg C) = 0.6$$
$$P(R \mid \neg H, C) = 0.7, \qquad P(\neg R \mid \neg H, C) = 0.3$$
$$P(R \mid \neg H, \neg C) = 0.1, \qquad P(\neg R \mid \neg H, \neg C) = 0.9$$

C. Conditional probability of Wet Grass

$$P(W \mid R) = 0.8, \qquad P(\neg W \mid R) = 0.2$$
$$P(W \mid \neg R) = 0.1, \qquad P(\neg W \mid \neg R) = 0.9$$

D. Conditional probability of School Off

$$P(O \mid R) = 0.7, \qquad P(\neg O \mid R) = 0.3$$
$$P(O \mid \neg R) = 0.2, \qquad P(\neg O \mid \neg R) = 0.8$$

Solution

Probability of Humid

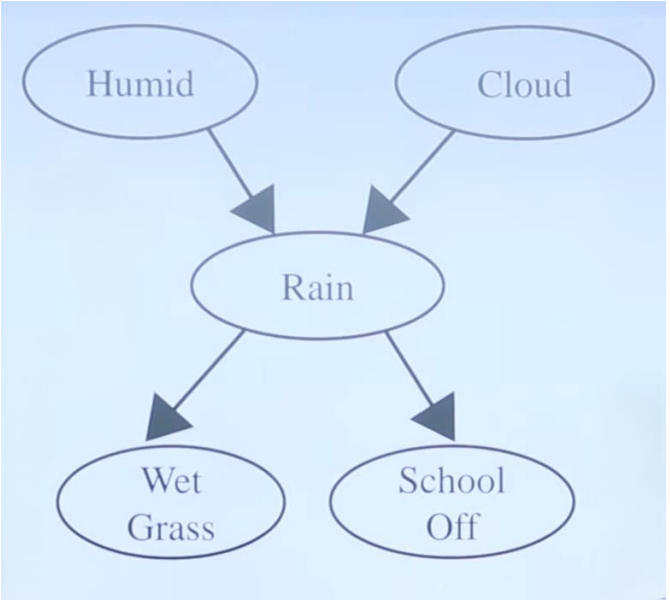
P(H)	$P(\neg H)$
0.2	0.8

Probability of Cloud

P(C)	$P(\neg C)$
0.6	0.4

Probability of Rain

H	C	$P(R)$	$P(\neg R)$
H	C	0.9	0.1
H	$\neg C$	0.4	0.6
$\neg H$	C	0.7	0.3
$\neg H$	$\neg C$	0.1	0.9



Probability of School off

R	P(O)	$P(\neg O)$
R	0.7	0.3
$\neg R$	0.2	0.8

Probability of Wet Grass

R	P(W)	$P(\neg W)$
R	0.8	0.2
$\neg R$	0.1	0.9

Solution

Probability that **school is off**, **it is raining**, **it is not humid**, and **it is cloudy,**

$$P(O, R, \neg H, C)$$

4) Formula used

Using the Bayesian network factorization:

$$P(O, R, \neg H, C) = P(\, O \mid R, \neg H, C\,) \cdot P(\, R \mid \neg H, C\,) \cdot P(\neg H) \cdot P(C)$$

Now, from the network, **School Off depends only on Rain**, so:

$$P(O \mid R, \neg H, C) = P(O \mid R)$$

Therefore,

$$P(O, R, \neg H, C) = P(O \mid R) \cdot P(R \mid \neg H, C) \cdot P(\neg H) \cdot P(C)$$

5) Substitute the values

From the tables:

$$P(\, O \mid R\,) = 0.7$$
$$(\, R \mid \neg H, C\,) = 0.7$$
$$P(\neg H) = 0.8$$
$$P(C) = 0.6$$

So,

$$P(O, R, \neg H, C) = 0.7 \times 0.7 \times 0.8 \times 0.6$$

Solution

7) Final answer

$$P(O, R, \neg H, C) = 0.2352$$

or as a percentage:

$$0.2352 \times 100 = 23.52\%$$

So the probability is:

$$23.52\%$$

Because in the network:

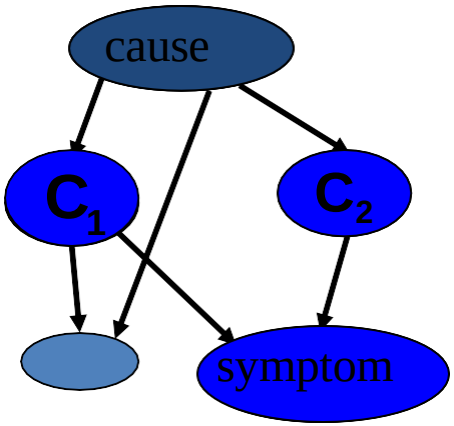
- H and C are root nodes
- R depends on H and C
- O depends only on R

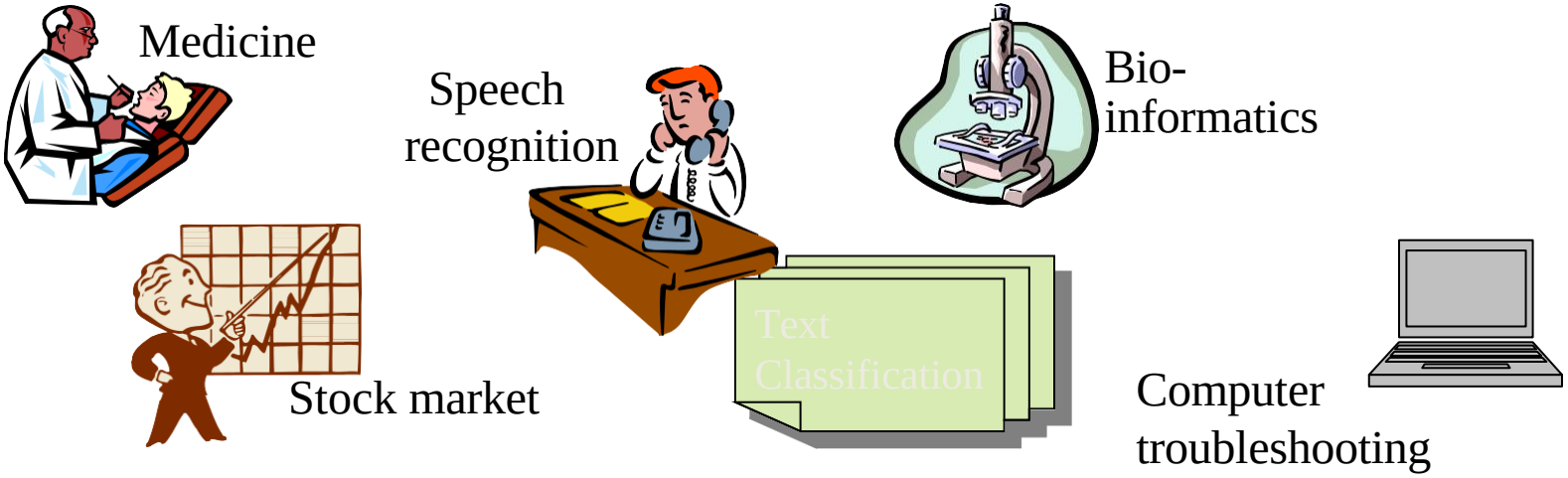
So the joint probability follows the arrows of the network:

$$P(O, R, \neg H, C) = P(O \mid R) \cdot P(R \mid \neg H, C) \cdot P(\neg H) \cdot P(C)$$

What Bayesian Networks are good for?

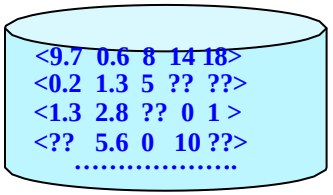
- Diagnosis: P(cause|symptom)=?
- Prediction: P(symptom|cause)=?
- Classification: $\max_{class} P(class|data)$
- Decision-making (given a cost function)

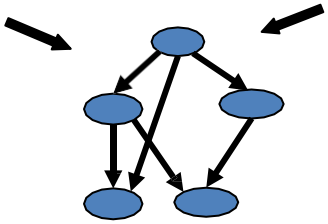




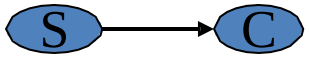
Why learn Bayesian networks?

- Combining domain expert knowledge with data





- Efficient representation and inference
- Incremental learning
- Handling missing data: **<1.3 2.8 ?? 0 1 >**
- Learning causal relationships:



THE END